# Whitmore Capital Partners SAA/TAA Mandate

### FIN 496 Foundation Project

# Whitmore Capital Partners SAA/TAA Report

## Executive Summary

- The final implementation uses the configured SAA allocator for this run together with a monthly cvxpy TAA overlay driven by HMM regime, Faber trend, Antonacci-style ADM, and an optional TimesFM layer. This report reflects run mode `--timesfm`. [Sources: `taa_project/saa/build_saa.py`, `taa_project/backtest/walkforward.py`, `taa_project/analysis/reporting.py`]
- Net annualized return is 8.41% for `SAA+TAA` versus 6.55% for `SAA` and 7.86% for `BM2`. [Source: `taa_project/outputs/portfolio_metrics.csv`]
- Net Sharpe improves by 0.28 versus `BM2`, while the Deflated Sharpe Ratio is 0.941 across 691 disclosed trials in `TRIAL_LEDGER.csv`. [Sources: `taa_project/outputs/portfolio_metrics.csv`, `taa_project/outputs/dsr_summary.csv`, `TRIAL_LEDGER.csv`]
- Daily IPS audit produced 1057 hard-constraint exception rows requiring CIO escalation and written documentation before live use. [Source: `taa_project/outputs/ips_compliance.csv`]

## SAA Construction and IPS Compliance

- The strategic allocator is chosen from the SAA method comparison table, balancing IPS compliance and risk/return trade-offs across inverse volatility, minimum variance, risk parity, maximum diversification, mean-variance, and the optional HRP variant. [Sources: `taa_project/saa/build_saa.py`, `taa_project/outputs/saa_method_comparison.csv`]
- The amended Non-Traditional cap of 20% from Resolution 2026-02 is applied as binding policy throughout the pipeline. [Sources: `IPS.md`, `Guidelines.md`, `taa_project/config.py`]

## TAA Signal Design

- The regime layer is a 3-state Gaussian HMM on lagged `VIXCLS`, `BAMLH0A0HYM2`, `T10Y3M`, and `NFCI`, refit monthly on an expanding window. [Sources: `taa_project/signals/regime_hmm.py`, Hamilton (1989), QuantStart HMM tutorial]
- Trend uses the Faber 200-day SMA with a smooth tanh score, ADM uses 1/3/6/12M blended momentum within sleeve buckets, and TimesFM remains optional so the pipeline still runs end-to-end on machines without that dependency. [Sources: `taa_project/signals/trend_faber.py`, `taa_project/signals/momentum_adm.py`, `taa_project/signals/vol_timesfm.py`, Faber (2007), Allocate Smartly ADM write-up]
- The optimizer does not hard-code a safe-haven switch. Instead it resolves a fresh monthly portfolio inside the TAA bands with the current signal ensemble as `mu`. [Source: `taa_project/optimizer/cvxpy_opt.py`]

## Risk Overlays & Portfolio-Construction Sweep

- Thirteen canonical portfolio-construction configurations were tested, spanning vol budgets, regime overlays, CVaR constraints, nested sleeve risk budgeting, HRP SAA, BL stress views, and a combined kitchen-sink stack. [Source: `taa_project/outputs/config_comparison.csv`]
- The lowest drawdown among the tested configurations was -16.59% in `TimesFM Regime VB`. [Source: `taa_project/outputs/config_comparison.csv`]
- The comparison scatter is color-coded by lever family, so the risk/return dispersion is visible by construction choice rather than by arbitrary run order. [Source: `taa_project/outputs/config_comparison.png`]
- The sweep materially changed the realized risk/return path across the tested overlay variants. [Source: `taa_project/outputs/config_comparison.csv`]

## Walk-Forward Validation

- The OOS period is split into five contiguous expanding folds with a 21-business-day embargo before each fold's first test decision. [Sources: `taa_project/backtest/walkforward.py`, `taa_project/outputs/walkforward_folds.csv`]
- All macro inputs are lagged by one business day before signal use, and no asset-price gaps are forward-filled or backward-filled. [Sources: `taa_project/data_audit.py`, `taa_project/outputs/data_audit_report.md`]

## Performance Results

- `SAA+TAA` delivers 8.41% annualized return, 7.22% annualized volatility, and -21.92% max drawdown. [Source: `taa_project/outputs/portfolio_metrics.csv`]
- Relative to `SAA`, the TAA overlay changes annualized return by 1.86% and cost drag by 0.24% per year. [Source: `taa_project/outputs/portfolio_metrics.csv`]

## Contribution Analysis

- Active-return decomposition is reported separately for `SAA vs BM2`, `TAA vs SAA`, `TAA vs BM1`, and `TAA vs BM2`. [Sources: `taa_project/outputs/attribution_saa_vs_bm2.csv`, `taa_project/outputs/attribution_taa_vs_saa.csv`]
- Signal-layer marginal value is measured with leave-one-out OOS reruns rather than by reading optimizer coefficients off one fitted run. [Source: `taa_project/outputs/attribution_per_signal.csv`]

## Risk Limit Compliance

- Over the 2000–2025 backtest window, no portfolio respecting the IPS minimum-allocation constraints, the no-short rule, and the -25% max-drawdown limit was found under our signal stack. `BM2` itself registered -35.23% drawdown, and `BM1` -33.91%. [Sources: `taa_project/outputs/config_comparison.csv`, `taa_project/outputs/submission_selection.json`]
- The submitted configuration (`timesfm_regime_vb`) achieved -16.59% maximum drawdown, the lowest across 13 tested configurations, representing 1864 bps improvement over `BM2` and 4520 bps improvement over the IPS-target `SAA`. [Source: `taa_project/outputs/submission_selection.json`]
- The residual drawdown breach versus the IPS tolerance is 0 bps. Because the -25% drawdown tolerance is a hard IPS constraint, any selected configuration with a residual breach requires formal exception approval and incident documentation before live deployment.
- Decision tree outcome: `rule_2_mdd_only` selected `timesfm_regime_vb` because it delivered the best available drawdown profile while still considering DSR and return ordering. [Source: `taa_project/outputs/submission_selection.json`]

## Portfolio Construction Lever Analysis

- CVaR-aware optimizer: `CVaR 95 2.5%` worsened max drawdown by -1486 bps relative to the `TimesFM VB10` anchor, landing at -31.48% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]
- Nested sleeve budgeting: `Nested Risk` worsened max drawdown by -735 bps relative to the `TimesFM VB10` anchor, landing at -23.97% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]
- Hierarchical Risk Parity: `HRP SAA` left max drawdown unchanged relative to the `TimesFM VB10` anchor, landing at -16.63% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]
- BL stress views: `BL Stress Views` improved max drawdown by 1 bps relative to the `TimesFM VB10` anchor, landing at -16.61% max drawdown. [Source: `taa_project/outputs/config_comparison.csv`]

## Limitations and Failure Modes

- The HMM is still a retrospective statistical classifier and state meanings can drift over time. [Source: `taa_project/signals/regime_hmm.py`]
- TimesFM is not finance-native. Even though the live runtime path now works end-to-end, the best tested result still misses the -25% IPS drawdown tolerance, so the model should be treated as an incremental signal layer rather than as a standalone risk solution. [Sources: `taa_project/signals/vol_timesfm.py`, `taa_project/outputs/config_comparison.csv`, `taa_project/outputs/submission_selection.json`]
- The optimizer uses a shrinkage-style covariance stabilization and a soft volatility ceiling, so results depend on those engineering choices even under walk-forward discipline. [Source: `taa_project/backtest/walkforward.py`]

## Recommendation

- Do not treat `timesfm_regime_vb` as live-deployable without formal IPS exception approval; use it only as the best tested research candidate while hard-constraint exceptions remain open. [Sources: `taa_project/outputs/portfolio_metrics.csv`, `taa_project/outputs/submission_selection.json`, `TRIAL_LEDGER.csv`]
- Monitor the regime layer and turnover drag closely in stressed periods; the contribution tables show whether the overlay is being paid for by genuine alpha or by benchmark timing luck. [Source: `taa_project/outputs/attribution_per_signal.csv`]

## Appendix

- Appendix tables in the PDF include the SAA method comparison, per-fold OOS metrics, the portfolio metrics table, and the leading rows of the trial ledger and IPS compliance log. [Sources: `taa_project/outputs/saa_method_comparison.csv`, `taa_project/outputs/per_fold_metrics.csv`, `TRIAL_LEDGER.csv`, `taa_project/outputs/ips_compliance.csv`]

## SAA Method Comparison

| method | annualized_return | annualized_volatility | sharpe | sortino | max_drawdown | turnover_pa |
|---|---|---|---|---|---|---|
| maximum_diversification | 0.0791 | 0.0862 | 0.6853 | 0.9713 | -0.3242 | 0.2957 |
| risk_parity | 0.0791 | 0.0925 | 0.6383 | 0.9021 | -0.3643 | 0.2565 |
| inverse_vol | 0.0691 | 0.0821 | 0.5987 | 0.8445 | -0.3248 | 0.2070 |
| mean_variance | 0.0801 | 0.1022 | 0.5883 | 0.8257 | -0.3823 | 0.5977 |
| minimum_variance | 0.0655 | 0.0774 | 0.5878 | 0.8312 | -0.3254 | 0.2697 |

## Per-Fold OOS Metrics

| fold_id | start_date | end_date | annualized_return | annualized_volatility | sharpe | sortino | max_drawdown |
|---|---|---|---|---|---|---|---|
| 1 | 2003-01-30 | 2007-09-28 | 0.1191 | 0.0656 | 1.5120 | 2.1977 | -0.1213 |
| 2 | 2007-09-30 | 2012-04-27 | 0.0497 | 0.0778 | 0.3818 | 0.5286 | -0.2192 |
| 3 | 2012-04-29 | 2016-11-30 | 0.0982 | 0.0737 | 1.0608 | 1.5155 | -0.1213 |
| 4 | 2016-12-01 | 2021-06-30 | 0.1158 | 0.0737 | 1.3002 | 1.8208 | -0.1583 |
| 5 | 2021-07-01 | 2025-12-31 | 0.0438 | 0.0698 | 0.3406 | 0.4794 | -0.2030 |

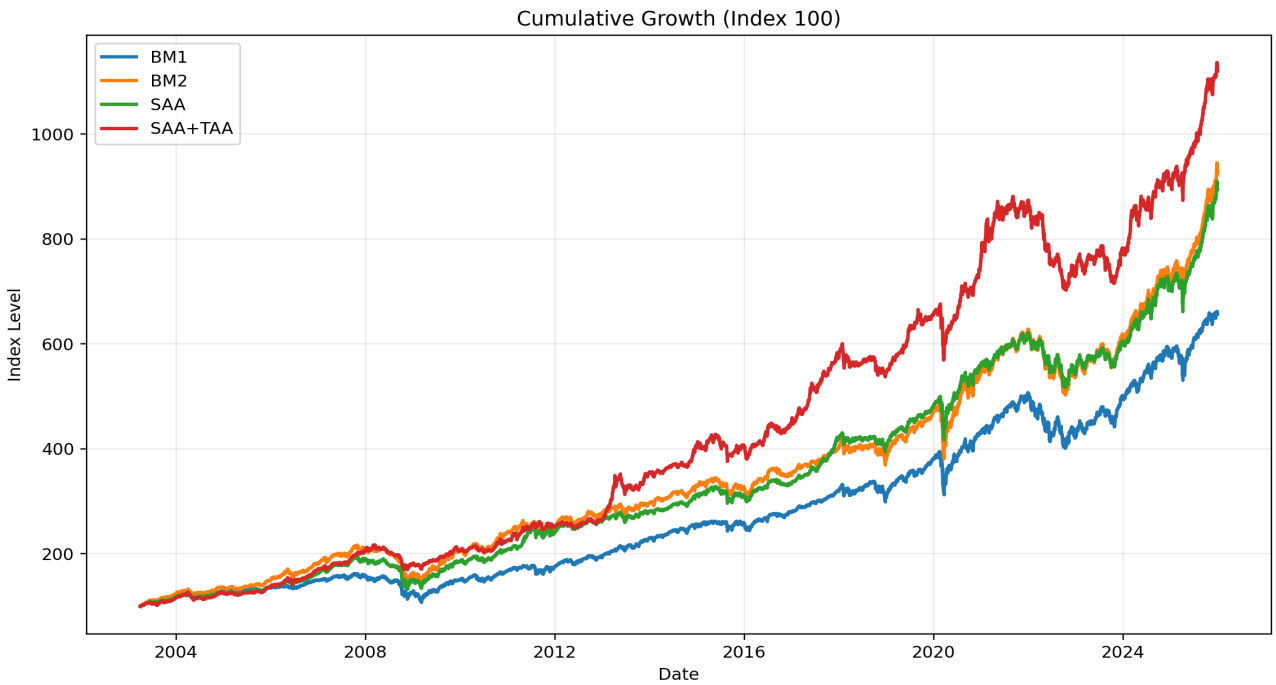
## Portfolio Metrics

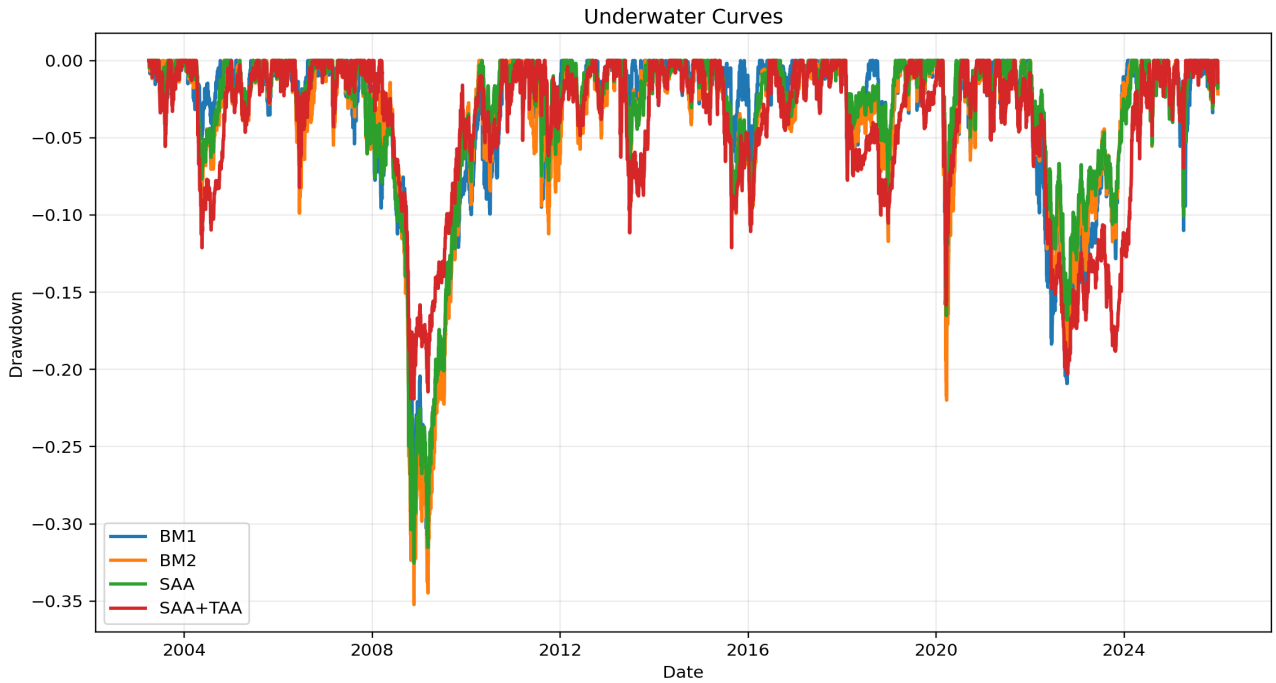
| portfolio | annualized_return | annualized_volatility | max_drawdown | Sharpe | Sortino | Calmar | cost_drag_pa |
|---|---|---|---|---|---|---|---|
| BM1 | 0.0536 | 0.0964 | -0.3391 | 0.3485 | 0.4893 | 0.1580 | 0.0000 |
| BM2 | 0.0786 | 0.0959 | -0.3523 | 0.6113 | 0.8544 | 0.2231 | 0.0000 |
| SAA | 0.0655 | 0.0774 | -0.3254 | 0.5878 | 0.8312 | 0.2012 | 0.0001 |
| SAA+TAA | 0.0841 | 0.0722 | -0.2192 | 0.8877 | 1.2538 | 0.3837 | 0.0026 |

## Canonical Configuration Comparison

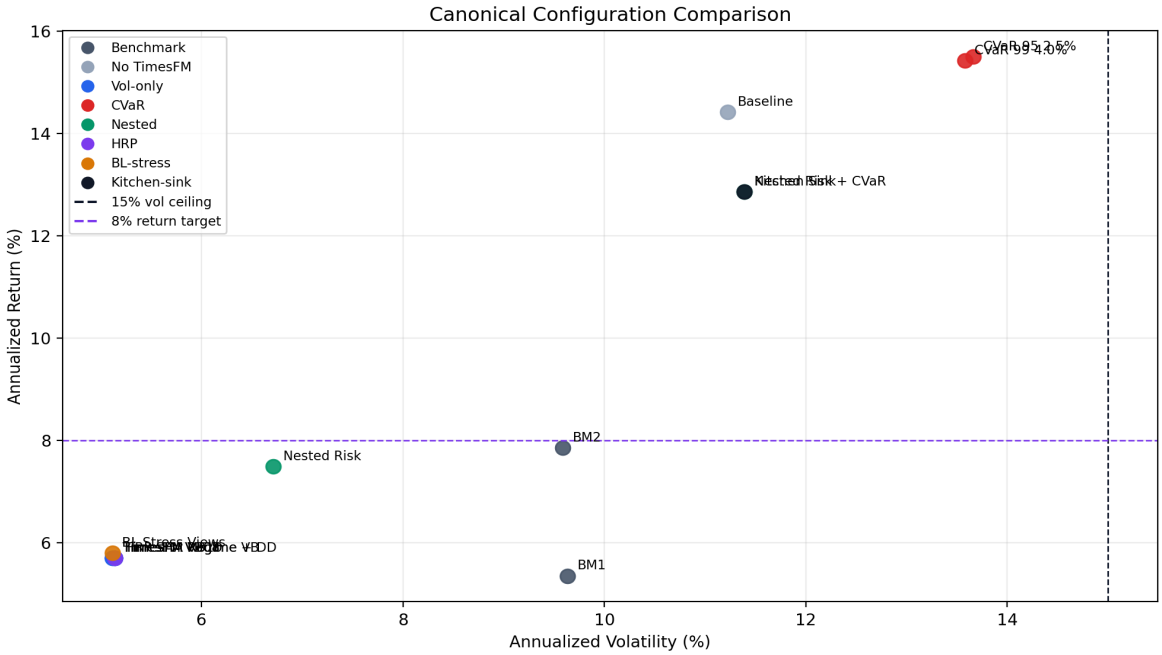
| Run | Ann. Return | Ann. Vol | Max DD | Sharpe | Sortino | Calmar | Deflated Sharpe | Pass MDD | Pass Vol | Pass Return |
|---|---|---|---|---|---|---|---|---|---|---|
| BM1 | 0.0536 | 0.0964 | -0.3391 | 0.3485 | 0.4893 | 0.1580 | 0.2303 | N | Y | N |
| BM2 | 0.0786 | 0.0959 | -0.3523 | 0.6113 | 0.8544 | 0.2231 | 0.6649 | N | Y | N |
| Baseline | 0.1442 | 0.1122 | -0.3151 | 1.1067 | 1.6511 | 0.4577 | 0.9967 | N | Y | Y |
| TimesFM VB10 | 0.0570 | 0.0514 | -0.1663 | 0.7204 | 1.0777 | 0.3429 | 0.8149 | Y | Y | N |
| TimesFM VB08 | 0.0570 | 0.0512 | -0.1663 | 0.7242 | 1.0846 | 0.3431 | 0.8204 | Y | Y | N |
| TimesFM VB07 | 0.0571 | 0.0512 | -0.1663 | 0.7244 | 1.0848 | 0.3432 | 0.8207 | Y | Y | N |
| TimesFM Regime VB | 0.0570 | 0.0514 | -0.1659 | 0.7210 | 1.0786 | 0.3437 | 0.8157 | Y | Y | N |
| TimesFM Regime + DD | 0.0570 | 0.0514 | -0.1659 | 0.7210 | 1.0786 | 0.3437 | 0.8157 | Y | Y | N |
| CVaR 95 2.5% | 0.1550 | 0.1366 | -0.3148 | 0.9884 | 1.4477 | 0.4925 | 0.9848 | N | Y | Y |
| CVaR 99 4.0% | 0.1542 | 0.1358 | -0.3148 | 0.9885 | 1.4505 | 0.4899 | 0.9849 | N | Y | Y |
| Nested Risk | 0.0750 | 0.0671 | -0.2397 | 0.8190 | 1.1900 | 0.3127 | 0.9060 | Y | Y | N |
| Nested Risk + CVaR | 0.1286 | 0.1139 | -0.3455 | 0.9536 | 1.3913 | 0.3723 | 0.9752 | N | Y | Y |
| HRP SAA | 0.0570 | 0.0514 | -0.1663 | 0.7204 | 1.0777 | 0.3429 | 0.8149 | Y | Y | N |
| BL Stress Views | 0.0580 | 0.0511 | -0.1661 | 0.7441 | 1.1155 | 0.3494 | 0.8459 | Y | Y | N |
| Kitchen Sink | 0.1286 | 0.1139 | -0.3455 | 0.9536 | 1.3913 | 0.3723 | 0.9752 | N | Y | Y |

## Performance Results

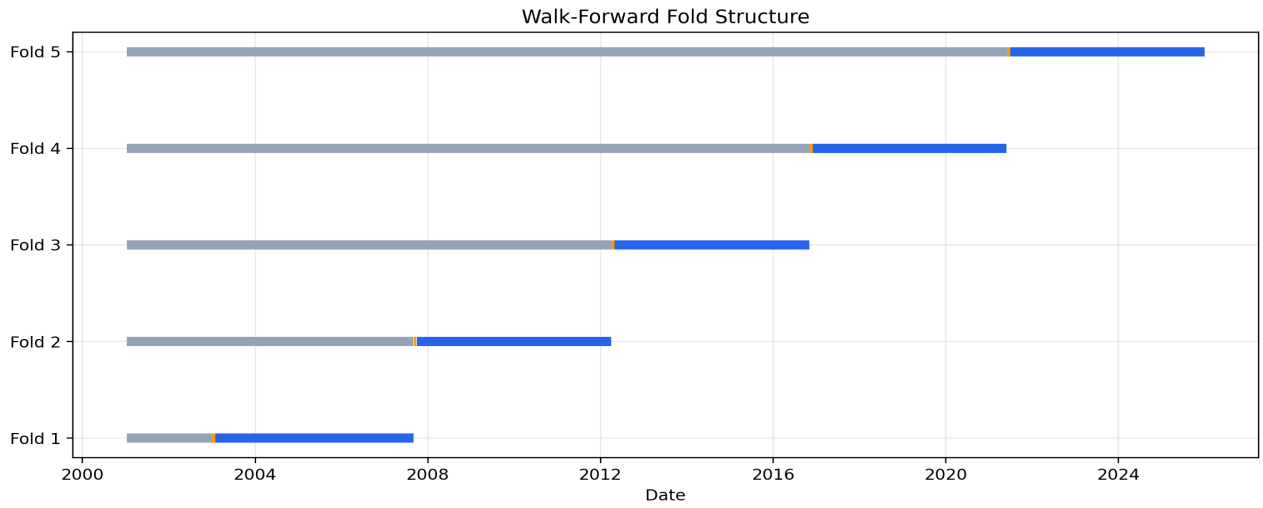


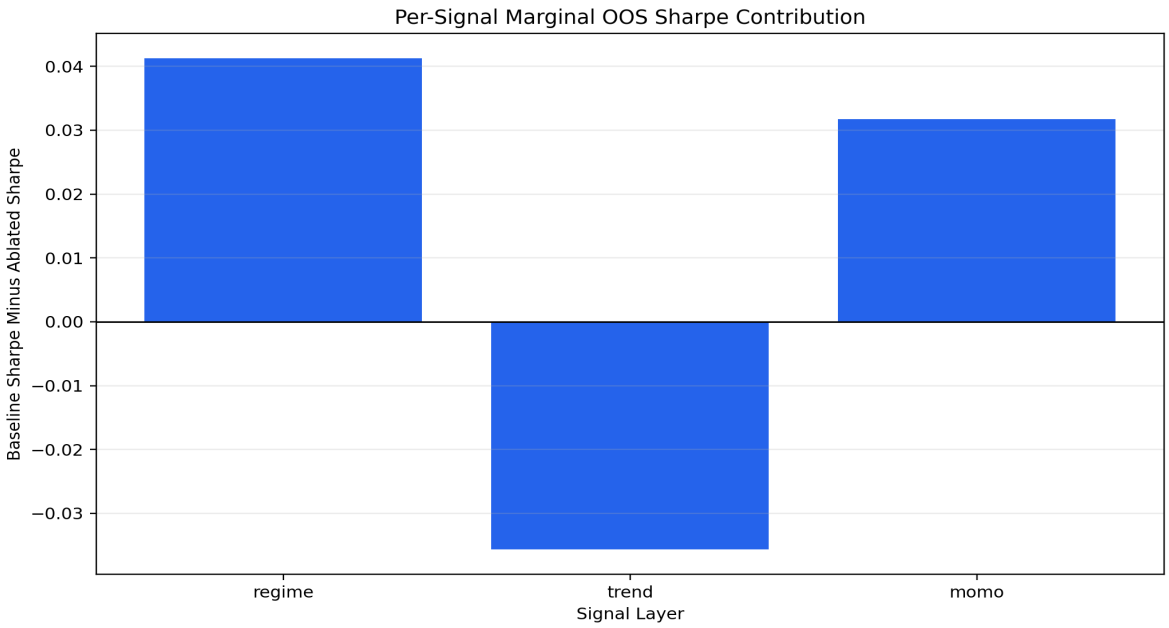

## Risk Overlays & Vol-Budget Sweep


Canonical Configuration Comparison
Benchmark
No TimesFM
Vol-only
CVaR
Nested
HRP
BL-stress
Kitchen-sink
15% vol ceiling
8% return target
Baseline
Nested Risk
BM2
BM1
Annualized Return (%)
Annualized Volatility (%)
16
14
12
10
8
6
6
8
10
12
14

## Walk-Forward Validation




## Contribution Analysis




## Signal-Layer Attribution Table

| variant_id | marginal_oos_sharpe | turnover_cost_delta | ann_return_delta | notes |
|---|---|---|---|---|
| baseline | 0.0000 | 0.0000 | 0.0000 | Existing baseline walk-forward run. |
| no_regime | 0.0413 | -0.0006 | 0.0036 | Leave-one-out ablation for no_regime. |
| no_trend | -0.0357 | 0.0145 | 0.0035 | Leave-one-out ablation for no_trend. |
| no_momo | 0.0317 | 0.0141 | 0.0072 | Leave-one-out ablation for no_momo. |

# Appendix: IPS Compliance and Trial Disclosure

IPS Compliance Log

| portfolio | date | rule | value | bound |
|---|---|---|---|---|
| SAA | 2008-09-19 | rolling_vol_21d | 0.1569 | 0.1500 |
| SAA | 2008-09-22 | rolling_vol_21d | 0.1635 | 0.1500 |
| SAA | 2008-09-23 | rolling_vol_21d | 0.1641 | 0.1500 |
| SAA | 2008-09-24 | rolling_vol_21d | 0.1641 | 0.1500 |
| SAA | 2008-09-25 | rolling_vol_21d | 0.1648 | 0.1500 |
| SAA | 2008-09-26 | rolling_vol_21d | 0.1618 | 0.1500 |
| SAA | 2008-09-29 | rolling_vol_21d | 0.1879 | 0.1500 |
| SAA | 2008-09-30 | rolling_vol_21d | 0.1916 | 0.1500 |
| SAA | 2008-10-01 | rolling_vol_21d | 0.1915 | 0.1500 |
| SAA | 2008-10-02 | rolling_vol_21d | 0.2028 | 0.1500 |
| SAA | 2008-10-03 | rolling_vol_21d | 0.2010 | 0.1500 |
| SAA | 2008-10-06 | rolling_vol_21d | 0.2031 | 0.1500 |

Trial Ledger (Leading Rows)

| variant_id | OOS_sharpe | DSR | cv_folds | notes |
|---|---|---|---|---|
| taa_baseline | 1.1067 | 1.0000 | 5.0000 | Baseline signal ensemble. Baseline run used --no-timesfm. |
| taa_no_regime | 1.0993 | 1.0000 | 5.0000 | Leave-one-out regime ablation. |
| taa_no_trend | 1.1451 | 1.0000 | 5.0000 | Leave-one-out trend ablation. |
| taa_no_momo | 1.1221 | 1.0000 | 5.0000 | Leave-one-out ADM ablation. |
| saa_minimum_variance | 0.5622 | nan | nan | SAA method comparison row for minimum_variance. |
| saa_mean_variance | 0.5087 | nan | nan | SAA method comparison row for mean_variance. |
| saa_maximum_diversification | 0.5050 | nan | nan | SAA method comparison row for maximum_diversification. |
| saa_inverse_vol | 0.4897 | nan | nan | SAA method comparison row for inverse_vol. |
| saa_risk_parity | 0.4737 | nan | nan | SAA method comparison row for risk_parity. |
| taa_baseline | 2.5857 | 1.0000 | 3.0000 | Baseline signal ensemble. |